Characterizing behavioral changes in energy use

IEOR 242 Final Project

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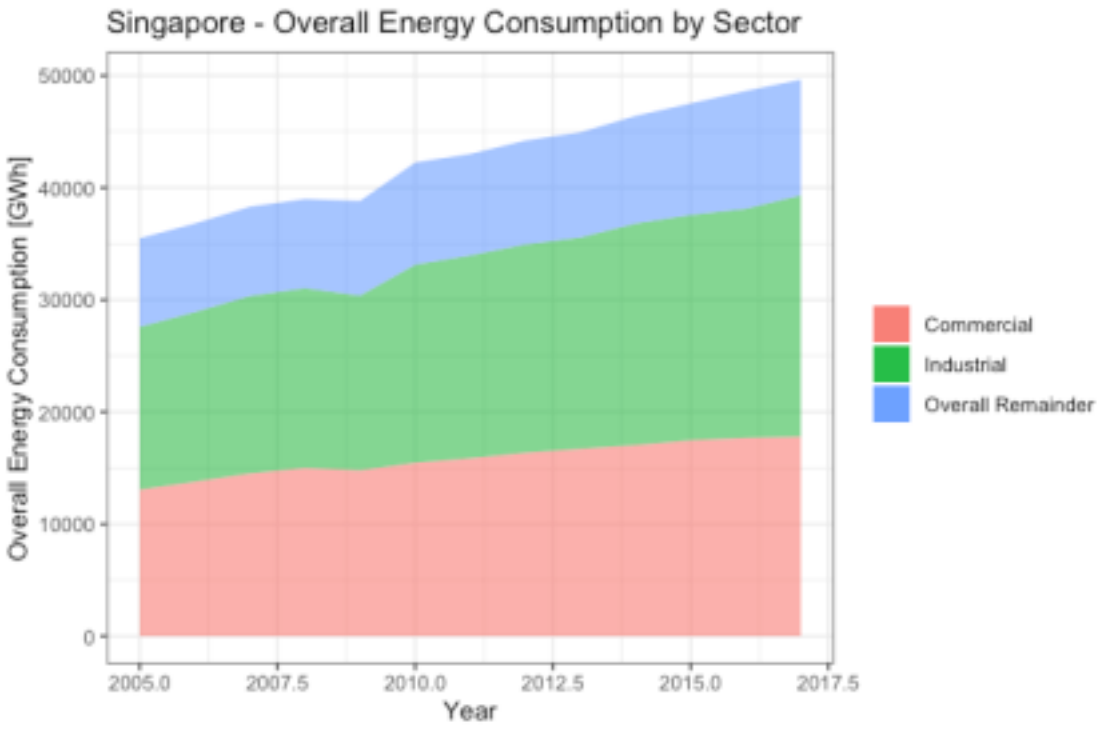
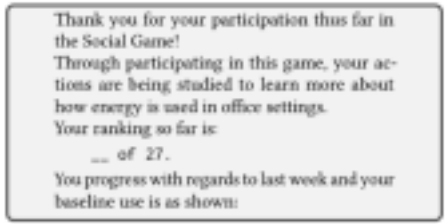
Motivation



Source: https://www.ema.gov.sg/Singapore_Energy_Statistics.aspx

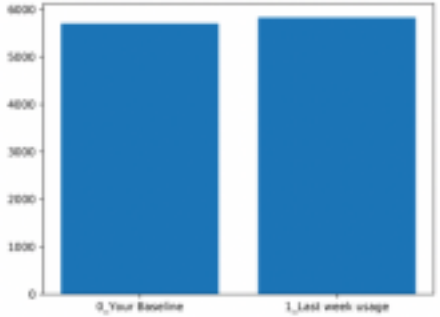
Research Experiment

Control


Thank you for your participation thus far in the Social Game!
Through participating in this game, your actions are being studied to learn more about how energy is used in office settings.
Your ranking so far is:
__ of 27.
You progress with regards to last week and your baseline use is as shown:

N/A

Treatment A: Engineering Visual



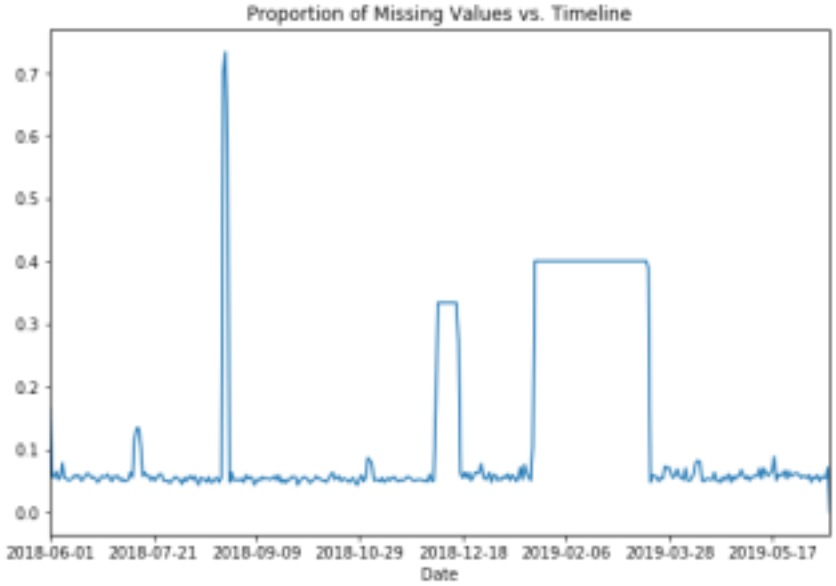
No statistical significance

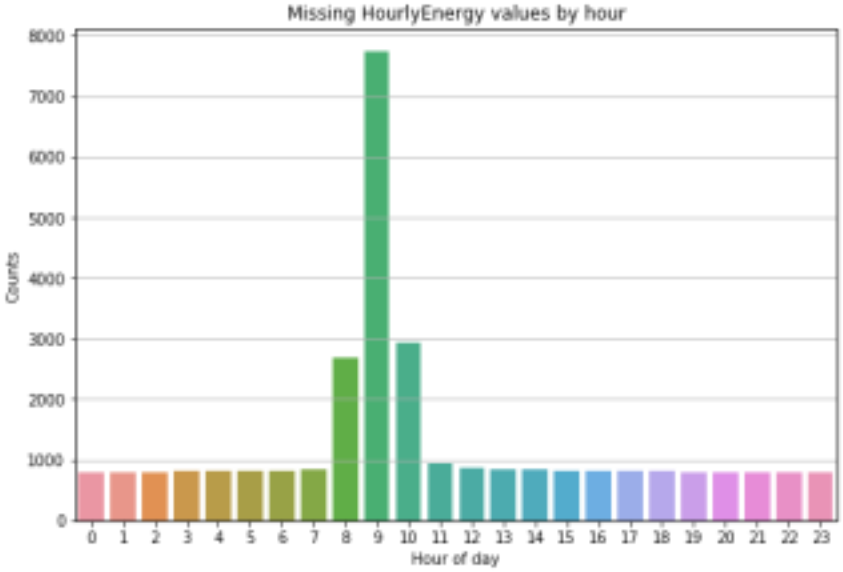
Treatment B: Ambient Visual

Avg. decrease of 82 Wh per day for three days following treatment

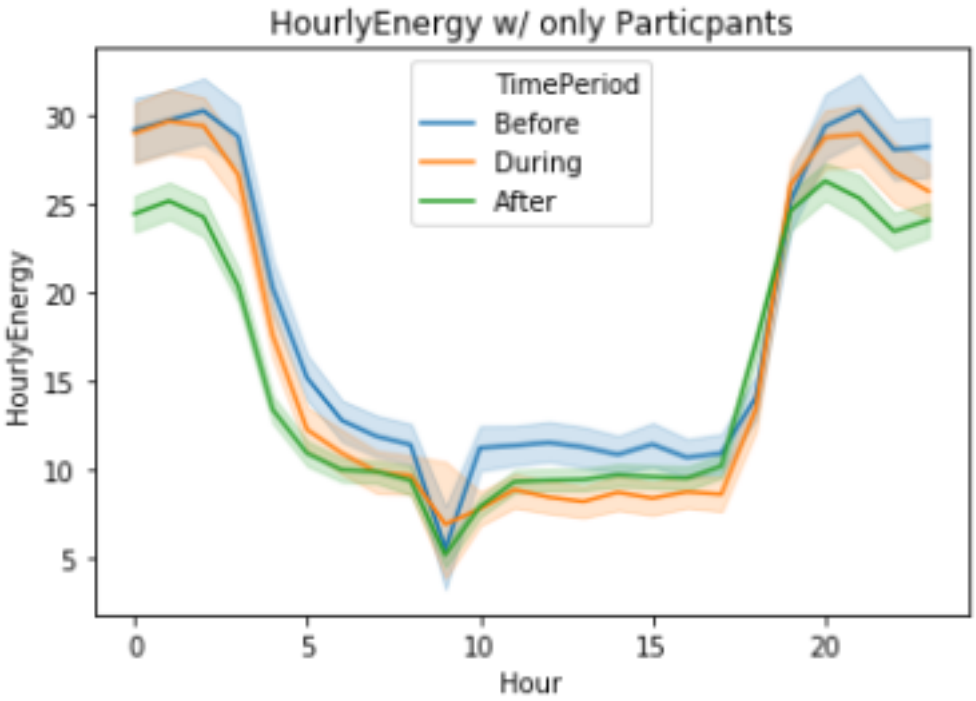
Source: “Engineering vs. Ambient type visualizations” (Spangher 2018)

Exploring the Data - N/A values

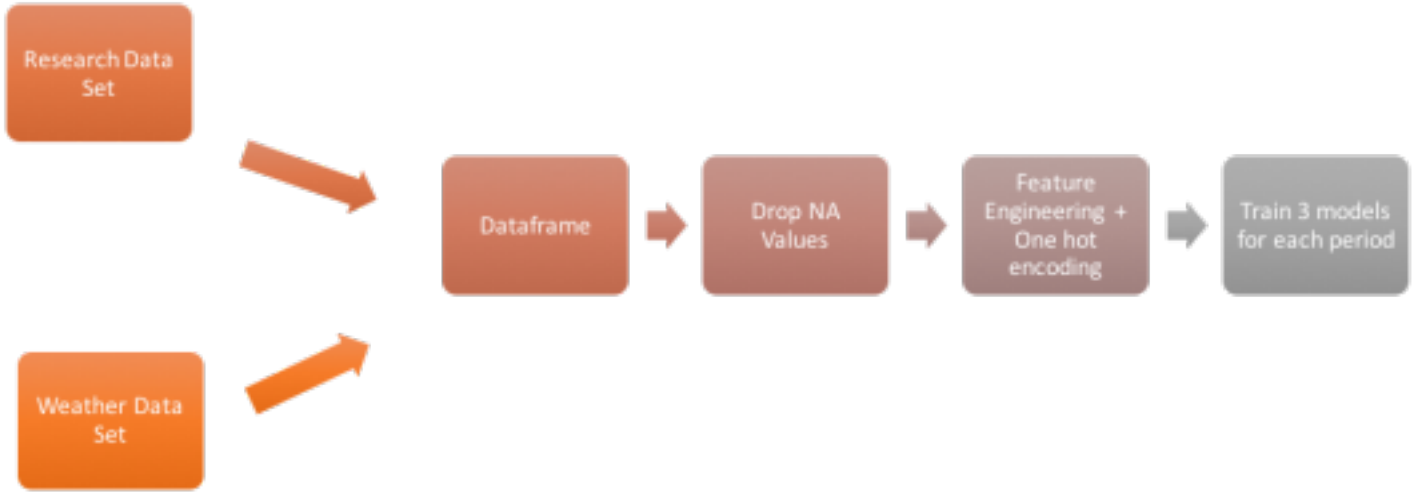




Exploring the Data - Hourly Energy Trends



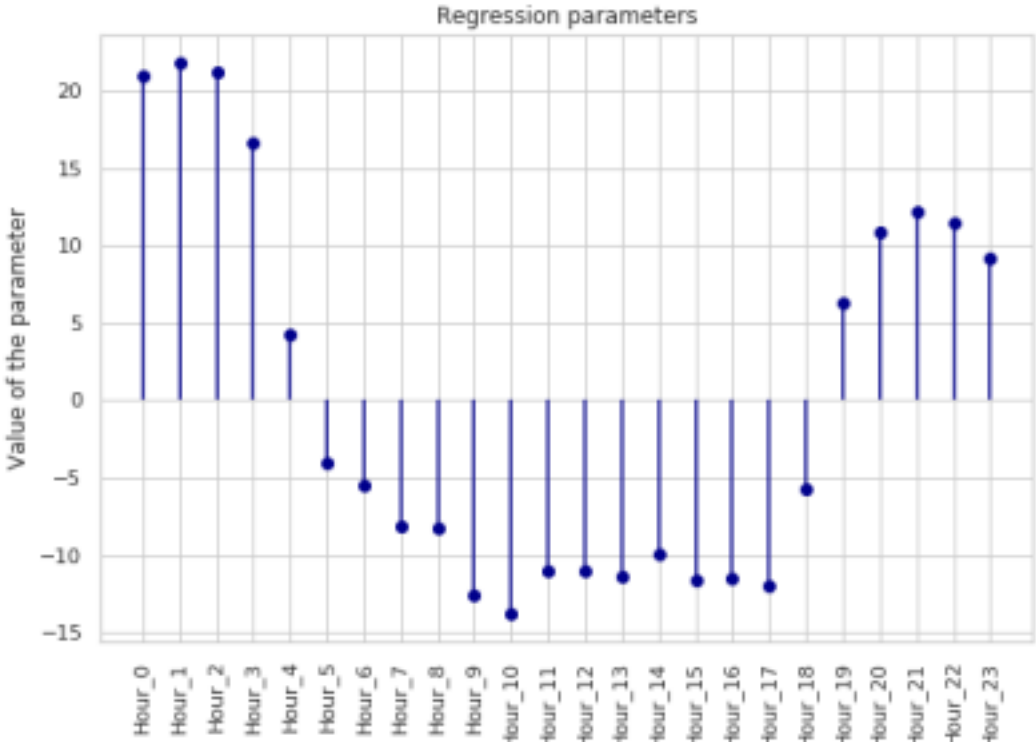
Data Transformation



Models

Linear regression:

- Interpretable
- Simple



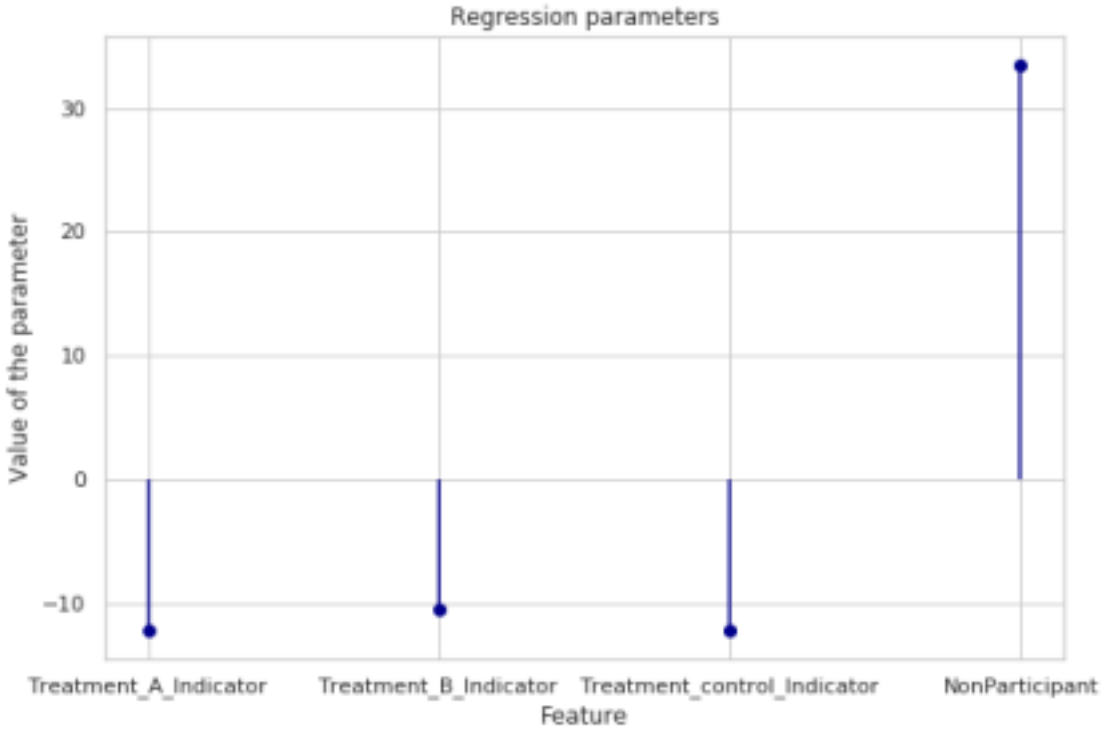
Models

Linear regression:

- Interpretable
- Simple

	Pre-treatment	During treatment	Post-treatment
R^2	0.508	0.685	0.494
MAE	16.02	18.79	44.31
RMSE	22.62	33.17	21.71

Models - Treatment Effect



Next steps

1. Include interaction terms in our regression models
2. Cluster hours of the day into groups or periods

Impact

- Scale learnings from the individual level to the aggregated level
- Inform future design of energy-saving gamification strategies
  - Identify the distribution of energy savings for targeted gamification strategies
  - Identify the post-treatment periods where the savings remain significant
- Use energy costs to determine optimal frequency of running the experiments for maximum savings

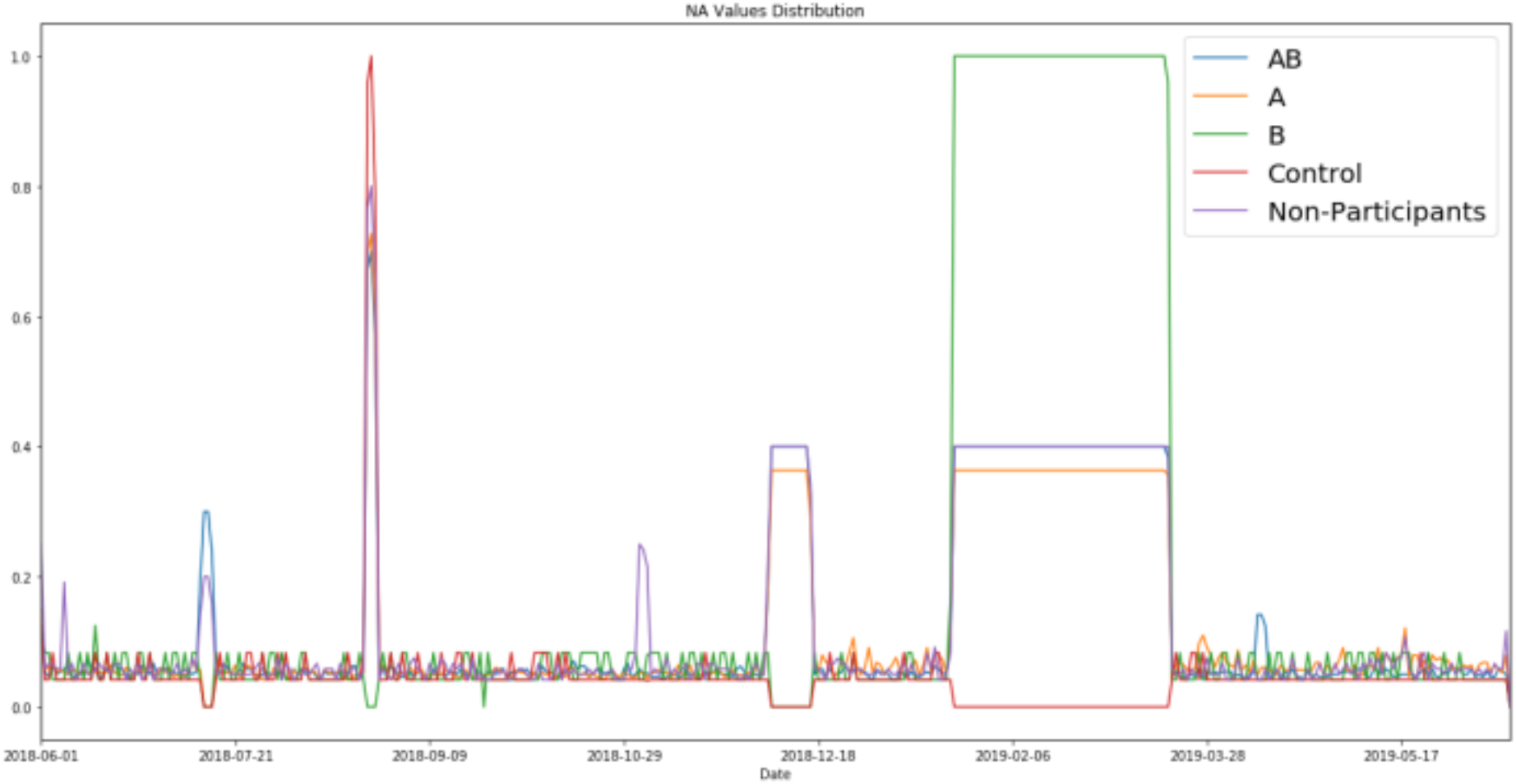
Q & A

Goals

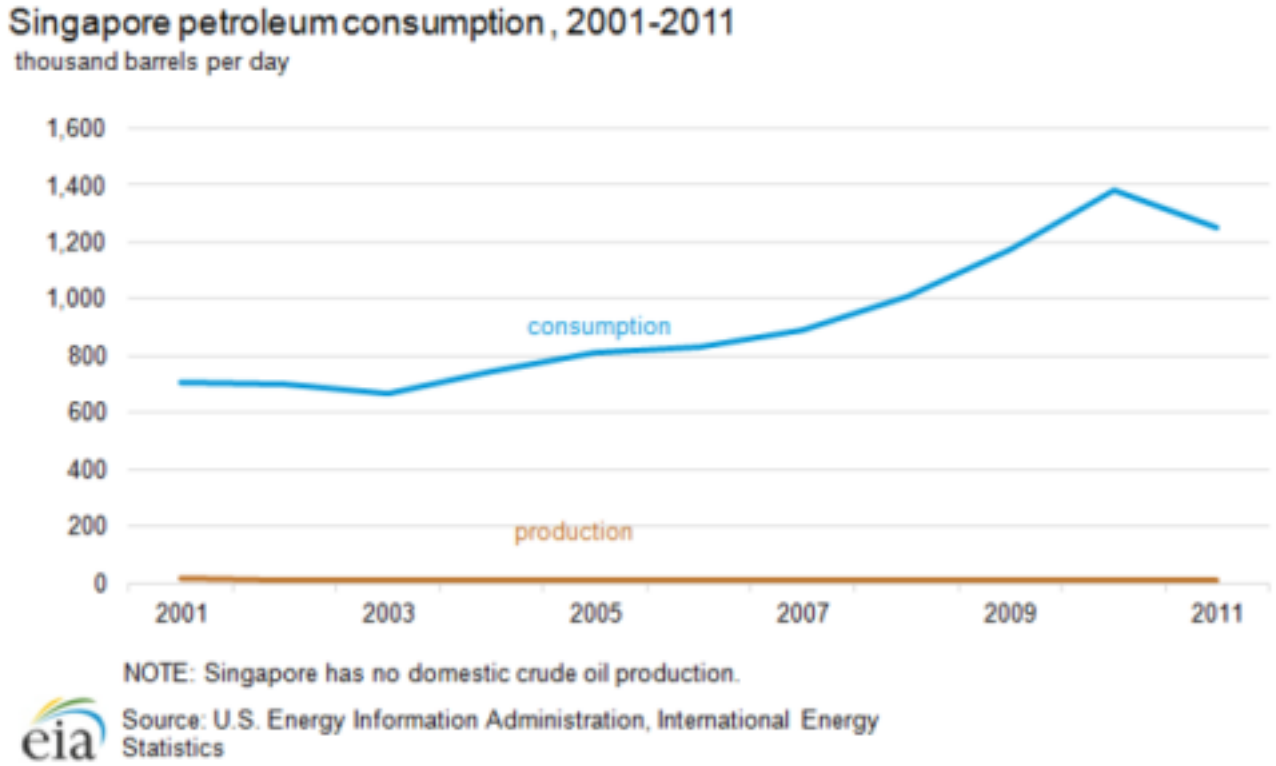
- Energy Impact
  - Does the experiment produce energy savings?
  - How are these energy savings distributed?
- User Impact
  - What is the range of user energy needs?
  - How are user behaviors affected during different time periods?
- What is the relationship between energy savings and treatments received over time?

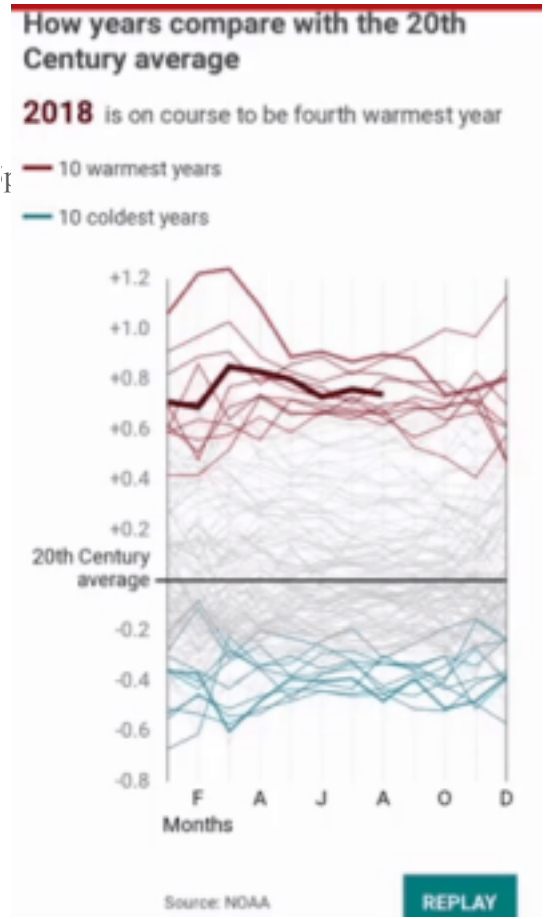
Name	Hourly Energy
Christopher	0.041000772
Keith	0.041110989
Ashish	0.04529924
Komang	0.047613799
Chayle	0.048164885
Sneha	0.049928359
Manish	0.051912267
Ramit	0.053124656
Giri	0.054777912

Name	Hourly Energy
Ola	0.193541276
Prakhar	0.193541276
Ajeet	0.19607627
Edwin	0.196296705
Vicky	0.198280613
Pandarasamy	0.20147691
Ken	0.205003858
Asit Mishra	0.222748815
Sivaraam	0.225063375


NA Values Distribution
1.0
0.8
0.6
0.4
0.2
0.0
2018-06-01
2018-07-21
2018-09-09
2018-10-29
2018-12-18
2019-02-06
2019-03-28
2019-05-17
Date
AB
A
B
Control
Non-Participants

Motivation


Singapore petroleum consumption, 2001-2011
thousand barrels per day
1,600
1,400
1,200
1,000
800
600
400
200
0
consumption
production
2001
2003
2005
2007
2009
2011
NOTE: Singapore has no domestic crude oil production.
eia
Source: U.S. Energy Information Administration, International Energy Statistics


How years compare with the 20th Century average
2018 is on course to be fourth warmest year
10 warmest years
10 coldest years
+1.2
+1.0
+0.8
+0.6
+0.4
+0.2
20th Century average
-0.2
-0.4
-0.6
-0.8
F
A
J
A
O
D
Months
Source: NOAA
REPLAY